Pre-analysis plan

We would like you to write a pre-analysis plan detailing the exact way how you will test each of the four hypotheses. In particular, for each hypothesis, please consider the following:

- Variables: Which variables will you include? How will these variables be coded - will you use them the way they are or recode them, and if so, how?
- Inclusion/exclusion criteria: Are there any observations you will exclude? If yes, which? Which years will you include/what period will you study?
- Exact analysis: Please write down the exact specification of the regression model, including the names of the dependent and independent variables, and whether you use robust standard errors or something else.
- Interpretation of results: Please specify which statistic you will take into account for testing the hypothesis. How do you interpret whether your test result is in line with the hypothesis or not?

There are some constraints on the planned analysis: The analysis has to be conducted as a regression analysis (not necessarily OLS), and there should be one regression table per hypothesis. In this table, there should be regression coefficients, standard errors, values of the corresponding test statistics, and p-values, as well as any additional hypothesis test results based on the regression equation.

H1. Having daughters leads to stronger support for gender equality (as measured by attitudes).

That is, denoting X as a measure of support for gender equality,

(X for respondents with daughters) > (X for respondents without daughters).

Variable type	Original variable name	Original variable codes	New variable name	New variable codes
Dependent	plj0621 plj0622	ordinal, 1-7 negative = missing	gendereq	factorial score higher values = higher gender equality
Independent	kidgeb1-19, kidmon1-19	birth year and month	ydaughters	continuous
Control	syear - gebjahr	continuous negative = missing	age	continuous
	pgbilzeit	continuous negative = missing	educ	continuous
	plh0182	0-10 negative = missing	lifesat	0-10
	syear	categorical: 2016, 2017, 2018	syear	categorical: 2016, 2017, 2018

Observations:

- All available years: 2016-2018; data is not panel
- All observations without missing values for original dependent variable

Analysis:

- Extended linear regression (eregress) - Basic linear regression with endogenous covariates
- Dependent variable: support for gender equality (as measured by attitudes); factorial score from items plj0621 (position of women: working as women) and plj0622 (position of women: working as married women)
- Independent variable: number of years with daughter(s)
- Control variables: male, age, education, life satisfaction, survey year
- Robust standard errors

Interpretation of results:

- Hypothesis testing: p-value associated to regression coefficient of independent variable
- The result is in line with hypothesis if $p \leq 0.05$

H2. Having daughters leads to more left-leaning political preferences.

That is, denoting Y as a measure of left-leaning political preferences,

(Y for respondents with daughters) > (Y for respondents without daughters).

Variable type	Original variable name	Original variable codes	New variable name	New variable codes
Dependent	plh0016 - plh0026	ordinal 1 - 5	left_att_x	factorial score higher values = higher left-leaning
Independent	kidgeb1-19, kidmon1-19	birth year and month	ydaughters	continuous
Control	syear - gebjahr	continuous negative = missing	age	continuous
	pgbilzeit	continuous negative = missing	educ	continuous
	plh0182	0-10 negative = missing	lifesat	0-10
	plh0007	1-4 negative = missing	polint	0-3

Observations:

- All available years (2002 and 2017)
- All observations with data for both years
- All observations without missing values for original dependent variables

Analysis:

- Fixed-effects linear regression (xtreg with option fe)
- Dependent variable: left-leaning attitudes (factorial score)
- Independent variable: number of years with daughter(s)
- Control variables: survey year (age), education, life satisfaction, political interest
- Robust standard errors

Interpretation of results:

- Hypothesis testing: p-value associated to regression coefficient of interaction term between independent variable and survey year
- The result is in line with hypothesis if $p \leq 0.05$

H3. Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).

That is, denoting Z as a measure of gender equality in the household,

(Z for respondents with daughters) > (Z for respondents without daughters).

pli0043_h – Hours Weekday Housework

Variable type	Original variable name	Original variable codes	New variable name	New variable codes
Dependent	pli0043_h Hours Weekday Housework	0-24 negative = missing	housework_equal	0-100 100 = perfect gender equality
Independent	kidgeb1-19, kidmon1-19	birth year and month	ydaughters	continuous
Control	syear - gebjahr	continuous negative = missing	age	continuous
	pgbilzeit	continuous negative = missing	educ	continuous
	pgfamstd	categorical negative = missing	married	0/1
	kidgeb1-19, kidmon1-19	birth year and month	kids	#
	pgemplst	categorical negative = missing	employed	0/1
	plh0182	0-10 negative = missing	lifesat	0-10

* Dependent and control variables were computed for both respondent and his/her partner.

** Stata formula for dependent variable:

housework_equal = 100 - (abs(hhousework - hhousework_p) / (hhousework + hhousework_p) * 100)

*** The model includes interaction between control variable for respondent and same control variable for his/her partner

Observations:

- All available years (2000 - 2020)
- All observations that have a partner (dyad observations)
- All observations without missing values for original dependent variables

Analysis:

- Fixed-effects linear regression (xtreg with option fe)
- Dependent variable: gender equality in the household (as measured by behaviors or outcomes); pli0043_h (Hours Weekday Housework)
- Independent variable: number of years with daughter(s)
- Control variables: survey year (age), education, life satisfaction, political interest
- Robust standard errors

Interpretation of results:

- Hypothesis testing: p-value associated to regression coefficient of interaction term between independent variable and survey year
- The result is in line with hypothesis if $p \leq 0.05$

H4. Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).

That is, denoting X as a measure of support for gender equality,

(X for male respondents with daughters − X for male respondents without daughters) >

(X for female respondents with daughters − X for female respondents without daughters)

Variable type	Original variable name	Original variable codes	New variable name	New variable codes
Dependent	plj0621 plj0622	ordinal, 1-7 negative = missing	gendereq	factorial score higher values = higher gender equality
Independent	kidgeb1-19, kidmon1-19	birth year and month	ydaughters	continuous
Control	syear - gebjahr	continuous negative = missing	age	continuous
	pgbilzeit	continuous negative = missing	educ	continuous
	plh0182	0-10 negative = missing	lifesat	0-10
	syear	categorical: 2016, 2017, 2018	syear	categorical: 2016, 2017, 2018
	sex	1=male, 2-woman	male	0/1

Observations:

- All available years: 2016-2018, but no panel data
- All observations without missing values for original dependent variables

Analysis:

- Extended linear regression (eregress) - Basic linear regression with endogenous covariates
- Dependent variable: support for gender equality (as measured by attitudes); factorial score from items plj0621 (position of women: working as women) and plj0622 (position of women: working as married women)
- Independent variable: number of years with daughter(s)
- Control variables: male, age, education, life satisfaction, survey year
- Robust standard errors

Interpretation of results:

- Hypothesis testing: p-value associated to regression coefficient of interaction term between independent variable and male
- The result is in line with hypothesis if $p \leq 0.05$